

Exercise I.

[20 marks]

Compute the following integrals:

1. $\int x^3 \ln x \, dx,$

3. $\int \frac{2x^2 + 1}{(x-1)(x^2+2)} \, dx,$

2. $\int \frac{(\ln x)^3 \ln(\ln x)}{x} \, dx,$

4. $\int \cos^2 x \sin^2 x \, dx.$

Exercise II.

[20 marks]

Let $\alpha \in \mathbb{R}$. Study the natures of the following series:

1. $\sum_{n=2}^{\infty} \frac{1}{(n-1)(n)},$

3. $\sum_{n=2}^{\infty} \arctan(\ln n),$

2. $\sum_{n=2}^{\infty} \frac{(\alpha-3)^n}{2^n n},$

4. $\sum_{n=2}^{\infty} \frac{1}{n e^n \ln n}.$

Exercise III.

[20 marks]

Let $\alpha \in \mathbb{R}$. Study the natures of the following improper integrals:

1. $\int_0^{\infty} e^{-x} \, dx,$

3. $\int_0^{\frac{\pi}{2}} \sqrt{\tan x} \, dx,$

2. $\int_0^{\infty} \frac{\sin x}{(1+x^{\frac{3}{2}}) \ln x} \, dx,$

4. $\int_0^1 \frac{1}{(1-x)^{\alpha} \sqrt[3]{\ln(1-x)}} \, dx.$

Exercise IV.

[30 marks]

Let $I_n = \frac{1}{n!} \int_0^1 x^n e^{-x} \, dx$ for all $n \in \mathbb{N}$.

1. Show that $\lim_{n \rightarrow \infty} I_n = 0.$

2. Find a relation between I_{n+1} and $I_n.$

3. Deduce that $\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{k!} = e.$

Exercise V.

[10 marks]

Let $g(x) = \int_{\ln(1+x)}^x \ln(e^t - 1) \, dt$. Compute $g'(x)$ and specify the domain of differentiability of g .

Exercise I. Compute the following integrals:

$$1. \int x^3 \ln x \, dx \stackrel{(1)}{=} \frac{x^4}{4} \ln x - \int \frac{x^4}{4} \cdot \frac{1}{x} \, dx \quad \begin{cases} u(x) \stackrel{(1)}{=} \ln x & v' = x^3 \\ u'(x) = \frac{1}{x} & v = \frac{x^4}{4} \end{cases}$$

$$\stackrel{(1)}{=} \frac{x^4}{4} \ln x - \frac{x^4}{16} + C.$$

$$2. \int \frac{(\ln x)^3 \ln(\ln x)}{x} \, dx \stackrel{(1)}{=} \int t^3 \ln t \, dt \quad \begin{cases} \text{put } \ln x = t, \frac{1}{x} dx = dt \end{cases}$$

$$\stackrel{\text{using 1.}}{=} \frac{t^4}{4} \ln t - \frac{t^4}{16} + C$$

$$\stackrel{(1)}{=} \frac{(\ln x)^4}{4} \ln(\ln x) - \frac{(\ln x)^4}{16} + C.$$

$$3. \int \frac{2x^2+1}{(x-1)(x^2+2)} \, dx \stackrel{(1)}{=} \int \frac{1}{x-1} \, dx + \int \frac{x+1}{x^2+2} \, dx$$

$$\stackrel{(1)}{=} \int \frac{1}{x-1} \, dx + \frac{1}{2} \int \frac{2x}{x^2+2} \, dx + \int \frac{1}{x^2+2} \, dx$$

$$\stackrel{(1)}{=} \ln|x-1| + \frac{1}{2} \ln(x^2+2) + \frac{1}{\sqrt{2}} \arctan \frac{x}{\sqrt{2}} + C$$

$$\begin{cases} \frac{2x^2+1}{(x-1)(x^2+2)} = \frac{a}{x-1} + \frac{bx+c}{x^2+2} \quad (1) \\ \Downarrow \\ 2x^2+1 = a(x^2+2) + (bx+c)(x-1) \\ \text{Now: } x=1 \Rightarrow \boxed{a=1} \quad (i) \\ x=0 \Rightarrow \boxed{c=1} \quad (ii) \\ x=-1 \Rightarrow \boxed{b=1} \quad (iii) \end{cases}$$

$$4. \int \cos^2 x \sin^2 x \, dx = \int \frac{1+\cos 2x}{2} \cdot \frac{1-\cos 2x}{2} \, dx = \frac{1}{4} \int (1-\cos^2 2x) \, dx$$

$$\stackrel{(1)}{=} \frac{1}{4} x - \int \frac{1+\cos 4x}{2} \, dx \stackrel{(1)}{=} \frac{1}{4} x - \frac{1}{2} x - \frac{1}{8} \sin 4x + C.$$

Exercise II. let $\alpha \in \mathbb{R}$. Study the nature of the following series:

$$1. \sum_{n=2}^{\infty} \frac{1}{(n-1)n}$$

Equivalent test: $\frac{1}{(n-1)n} \stackrel{(2)}{\sim} \frac{1}{n^2} > 0$, $\sum \frac{1}{n^2}$ conv (Riemann) $\Rightarrow \sum \frac{1}{(n-1)n}$ conv. (1)

Comparison test: $\frac{1}{(n-1)n} \leq \frac{1}{\frac{n}{2} \cdot n} = 2 \cdot \frac{1}{n^2}$, $\sum \frac{1}{n^2}$ conv $\Rightarrow \sum \frac{1}{(n-1)n}$ conv. (1)

Telescoping series: $\sum_{n=2}^{\infty} \frac{1}{(n-1)n} \stackrel{(2)}{=} \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) \stackrel{(1)}{=} \left(\lim_{n \rightarrow \infty} \frac{1}{n} - \frac{1}{2-1} \right) = 1$ conv. (1)

n^α -test: $n^2 \cdot \frac{1}{(n-1)n} \stackrel{(3)}{\rightarrow} 1 \neq 0$, $\alpha = 2 > 1 \Rightarrow \sum \frac{1}{(n-1)n}$ conv. (1)

or $n^{1.5} \cdot \frac{1}{(n-1)n} \stackrel{(3)}{\rightarrow} 0$, $\alpha = 1.5 > 1 \Rightarrow \sum \frac{1}{(n-1)n}$ conv. (1)

$$2. \sum \frac{(\alpha-3)^n}{2^n n}. \text{ let } u_n = \frac{(\alpha-3)^n}{2^n n}$$

$$\text{Ratio test: } \left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{(\alpha-3)^{n+1} / 2^{n+1} (n+1)}{(\alpha-3)^n / 2^n n} \right| = \frac{|\alpha-3| n}{2(n+1)} \xrightarrow{n \rightarrow \infty} \frac{1}{2} |\alpha-3|$$

$$\text{or Root test: } \lim_{n \rightarrow \infty} |u_n|^{1/n} = \lim_{n \rightarrow \infty} \frac{|\alpha-3|}{2 n^{1/n}} = \frac{1}{2} |\alpha-3|$$

i) If $\frac{1}{2} |\alpha-3| < 1$, i.e. $1 < \alpha < 5$, then $\sum u_n$ conv. absol. (1)

ii) If $\frac{1}{2} |\alpha-3| > 1$, i.e. $\alpha < 1$ or $\alpha > 5$, then $\sum u_n$ div. (1)

iii) If $\frac{1}{2} |\alpha-3| = 1$, i.e. $\alpha = 1$ or $\alpha = 5$, Tests give no info. (1)

Case $\alpha = 1$: $\sum u_n = \sum \frac{(-2)^n}{2^n n} = \sum \frac{(-1)^n}{n}$ conv. (Alternating series test) (1)

Case $\alpha = 5$: $\sum u_n = \sum \frac{2^n}{2^n n} = \sum \frac{1}{n}$ Divergent. (1) $\left\{ \begin{array}{l} \text{Series test} \\ \frac{1}{n} > 0, \downarrow, \rightarrow 0 \end{array} \right.$

$$3. \sum_{n=2}^{\infty} \arctan(\ln n), \quad \lim_{n \rightarrow \infty} \arctan(\ln n) = \frac{\pi}{2} \neq 0 \Rightarrow \text{Series div.} \quad (2)$$

$$4. \sum_{n=2}^{\infty} \frac{1}{n e^n \ln n}$$

Comparison test: $0 < \frac{1}{n e^n \ln n} < \frac{1}{e^n}$; $\sum \frac{1}{e^n}$ conv. (G.S.) $\Rightarrow$ Series conv. (1)

n^α test: $n^{10} \cdot \frac{1}{n e^n \ln n} \xrightarrow{n \rightarrow \infty} 0$, $\alpha = 10 > 1$ then the series conv. (1)

Exercise III. let $\alpha \in \mathbb{R}$. Study the natures of the following imp. integrals.

$$1. \int_0^{\infty} e^{-x} dx \stackrel{(1)}{=} \lim_{\alpha \rightarrow \infty} \int_0^{\alpha} e^{-x} dx \stackrel{(1)}{=} \lim_{\alpha \rightarrow \infty} [-e^{-x}]_0^{\alpha} \stackrel{(1)}{=} \lim_{\alpha \rightarrow \infty} [-e^{-\alpha} + 1] \stackrel{(1)}{=} 1 \text{ conv.}$$

or α^α test: $x^{10} e^{-x} \xrightarrow{x \rightarrow \infty} 0$, $\alpha = 10 > 1 \Rightarrow$ the imp. int. conv. (1)

$$2. \int_0^{\infty} \frac{\sin x}{(1+x^{3/2}) \ln x} dx \stackrel{(1)}{=} \int_0^1 + \int_1^2 + \int_2^{\infty} = I_1 + I_2 + I_3 \text{ formally.}$$

near $1^\pm$: $\frac{\sin x}{(1+x^{3/2}) \ln x} \sim \frac{\sin 1}{2(x-1)}$ has a sign $\Rightarrow I_1 \& I_2$ div. (1)
(by Riemann's test)

Thus the imp st diverges. ⁽¹⁾

near $+\infty$: $\left| \frac{\sin x}{(1+x^{3/2}) \ln x} \right| \leq \frac{1}{(1+x^{3/2}) \ln x} \sim \frac{1}{x^{3/2} \ln x}$ ^(1B) $\alpha = 3/2 > 1 \Rightarrow I_{\text{conv}}$

3. $\int_0^{\pi/2} \sqrt{t \tan x} \cdot dx$; $x = u(t) = \arctan t$, u st $\nearrow$ on $\mathbb{R}$
 $\frac{dx}{dt} = \frac{1}{1+t^2}$ ⁽¹⁾

⁽¹⁾ $= \int_0^\infty \sqrt{t} \cdot \frac{1}{1+t^2} dt$

$\frac{\sqrt{t}}{1+t^2} \sim \frac{t^{1/2}}{t^2} = \frac{1}{t^{3/2}} > 0$, $\int \frac{1}{t^{3/2}} dt \text{ conv} \Rightarrow \int_0^{\pi/2} \sqrt{t \tan x} \text{ conv.}$ ⁽¹⁾

4. $I_\alpha = \int_0^1 \frac{1}{(1-x)^\alpha \sqrt[3]{\ln(1-x)}} dx$ ⁽¹⁾ $= \int_0^{1/2} + \int_{1/2}^1 = I_{\alpha_1} + I_{\alpha_2}$

near 0^+ : $f(x) \sim_{0^+} \frac{1}{\sqrt[3]{-x}} = -\frac{1}{x^{1/3}} < 0 \Rightarrow I_{\alpha_1} \text{ conv (Riemann equiv test)}$ ⁽¹⁾

near 1^- : $f(x) \sim_{1^-} \frac{1}{(1-x)^\alpha [\ln(1-x)]^{1/3}} < 0 \Rightarrow I_{\alpha_2} \text{ conv iff } \alpha < 1$ ^(1B)
 (Bertrand + change of variable)

Thus $I_\alpha \text{ conv iff } I_{\alpha_1} \& I_{\alpha_2} \text{ both conv iff } \alpha < 1$. ^(1B)

Exercise IV. Let $I_n = \frac{1}{n!} \int_0^1 x^n e^{-x} dx \quad \forall n \in \mathbb{N}$.

1. Show that $\lim_{n \rightarrow \infty} I_n = 0$.

1st way: $|I_n| \leq \frac{1}{n!} \underbrace{\int_0^1 e^{-x} dx}_{\in \mathbb{R}} \xrightarrow[n \rightarrow \infty]{} 0 \Rightarrow I_n \xrightarrow[n \rightarrow \infty]{} 0$ ⁽⁵⁾ ⁽²⁾ ⁽³⁾

2nd way, MVT: $I_n = \frac{1}{n!} \int_0^1 \underbrace{x^n}_g \underbrace{e^{-x}}_f dx$ ⁽³⁾ $\begin{cases} \cdot f \in C([0,1]) \text{ (1)} \\ \cdot g \geq 0 \text{ on } [0,1] \text{ (1)} \\ \cdot 0 < \xi \leq 1 \text{ (1)} \end{cases}$
 $\stackrel{\text{M.V.T}}{=} e^{-\xi} \frac{1}{n!} \int_0^1 x^n dx$
 $\stackrel{(1)}{=} e^{-\xi} \frac{1}{(n+1)!} \xrightarrow[n \rightarrow \infty]{} 0$ ⁽¹⁾

or
$$I_n = \frac{1}{n!} \int_0^1 \underbrace{x^n}_f \underbrace{e^{-x}}_g dx$$

MVT
$$= \frac{1}{n!} \xi^n \int_0^1 e^{-x} dx$$

$$= \underbrace{\frac{1}{n!}}_{\downarrow 0} \underbrace{\xi^n [1 - e^{-1}]}_{\text{bdd}} \xrightarrow{n \rightarrow \infty} 0$$

$$\begin{cases} \bullet f \in C([0,1]) \\ \bullet g \geq 0 \text{ on } [0,1] \\ \bullet 0 \leq \xi \leq 1 \end{cases}$$

2. Find a relation between I_{n+1} and I_n .

$$I_{n+1} = \frac{1}{(n+1)!} \int_0^1 \underbrace{x^{n+1}}_u \underbrace{e^{-x}}_{v'} dx$$

$$= \frac{1}{(n+1)!} \left[\underbrace{-x^{n+1} e^{-x}}_{\text{v} \cdot \text{e}^{-1}} \Big|_0^1 + (n+1) \int_0^1 \underbrace{x^n e^{-x}}_{\text{v}} dx \right]$$

$$= \frac{-e^{-1}}{(n+1)!} + I_n$$

or
$$I_n = \frac{1}{n!} \int_0^1 \underbrace{x^n}_{v'} \underbrace{e^{-x}}_u dx$$

$$= \frac{1}{n!} \left[\frac{x^{n+1}}{n+1} e^{-x} \Big|_0^1 + \frac{1}{n+1} \int_0^1 x^{n+1} e^{-x} dx \right]$$

$$= \frac{e^{-1}}{(n+1)!} + I_{n+1}$$

3. Deduce that $\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{k!} = e$.

$$\sum_{k=0}^n \frac{1}{k!} \stackrel{(2)}{=} 1 + \sum_{k=1}^n \frac{1}{k!} \stackrel{(2)}{=} 1 + \sum_{k=1}^n \frac{(I_k - I_{k-1})}{-e^{-1}}$$

$$\stackrel{(3)}{=} 1 - e [I_n - I_0] \xrightarrow{n \rightarrow \infty} 1 + e I_0 \stackrel{(2)}{=} 1 + e \int_0^1 e^{-x} dx$$

$$= 1 - e e^{-x} \Big|_0^1 \stackrel{(1)}{=} 1 - e [e^{-1} - 1] = e$$

Exercise V. Let $g(x) = \int^x \ln(e^t - 1) dt$. Compute $g'(x)$ and specify the domain of diff of g .

$$\bullet g'(x) = u'(x) f(u(x)) - v'(x) f(v(x)) \quad (2)$$

$$= \ln(e^x - 1) - \frac{1}{1+x} \ln x.$$

• Domain of diff of g :

i) J , Domain of cont of f ; $e^t - 1 > 0 \Leftrightarrow e^t > 1 \Leftrightarrow t > 0$

So $J =]0, \infty[$ (1)

ii) u, v diff on $\mathbb{R} \cap]-1, +\infty[=]-1, +\infty[= I_1 \cap I_2$ (1)

$u(x) \in J \Leftrightarrow x \in]0, +\infty[= I_3$ (1)

$v(x) \in J \Leftrightarrow \ln(1+x) > 0 \Leftrightarrow 1+x > 1 \Leftrightarrow x \in]0, +\infty[= I_4$ (1)

Thus $I = I_1 \cap I_2 \cap I_3 \cap I_4 =]0, +\infty[$ is the domain of diff of g .